Programmable Car

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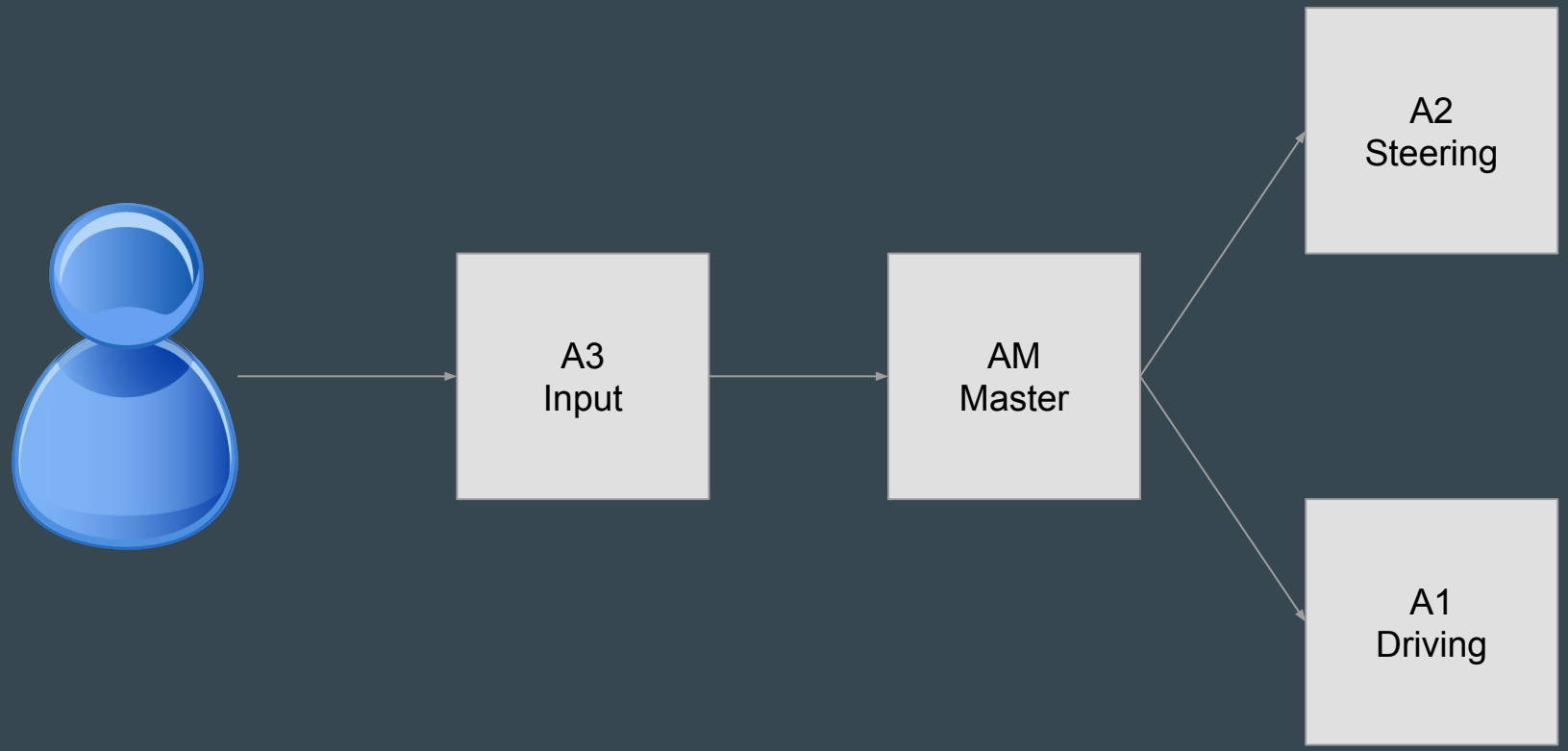
Programmable Car - Abstract

This project is a programmable LEGO toy car that follows a path set by a user. Utilizing an onboard interface, users input up to ten directional moves via buttons and select speeds using a potentiometer. Inside, four Arduinos work together using our own custom JSON communication code. One Arduino acts as the main brain, while the others control the driving, steering, and an LCD screen. The car also has an ultrasonic sensor to avoid crashing into obstacles. Overall, it is an original project that shows how multiple Arduinos can connect and share tasks.

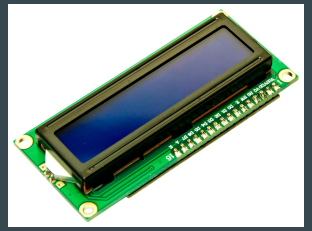
Project Idea

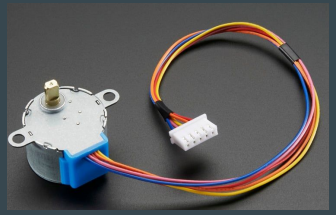
- This project is a programmable toy car that executes a user-defined sequence of up to ten cardinal-direction moves.
- In “reading mode,” users enter moves using four directional buttons and select one of three speeds (slow, moderate, fast) with a potentiometer.
- The car switches to “driving mode” when the “GO” button is pressed or the move limit is reached.
- An onboard ultrasonic sensor provides automatic collision avoidance.
- The car’s shell and components (e.g., wheels) are built from LEGO pieces.

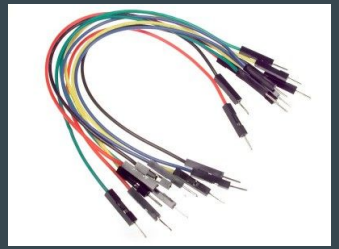
Role of Each Arduino - AM, A1, A2, A3

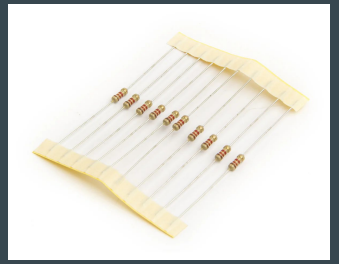


IO Device Types Used



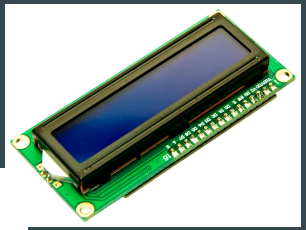




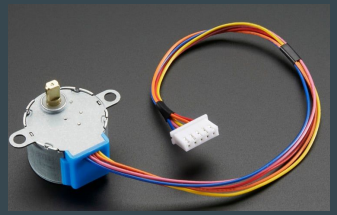


IO Device Implementation

A3

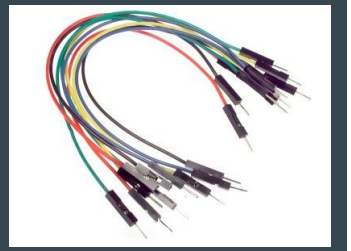


A1



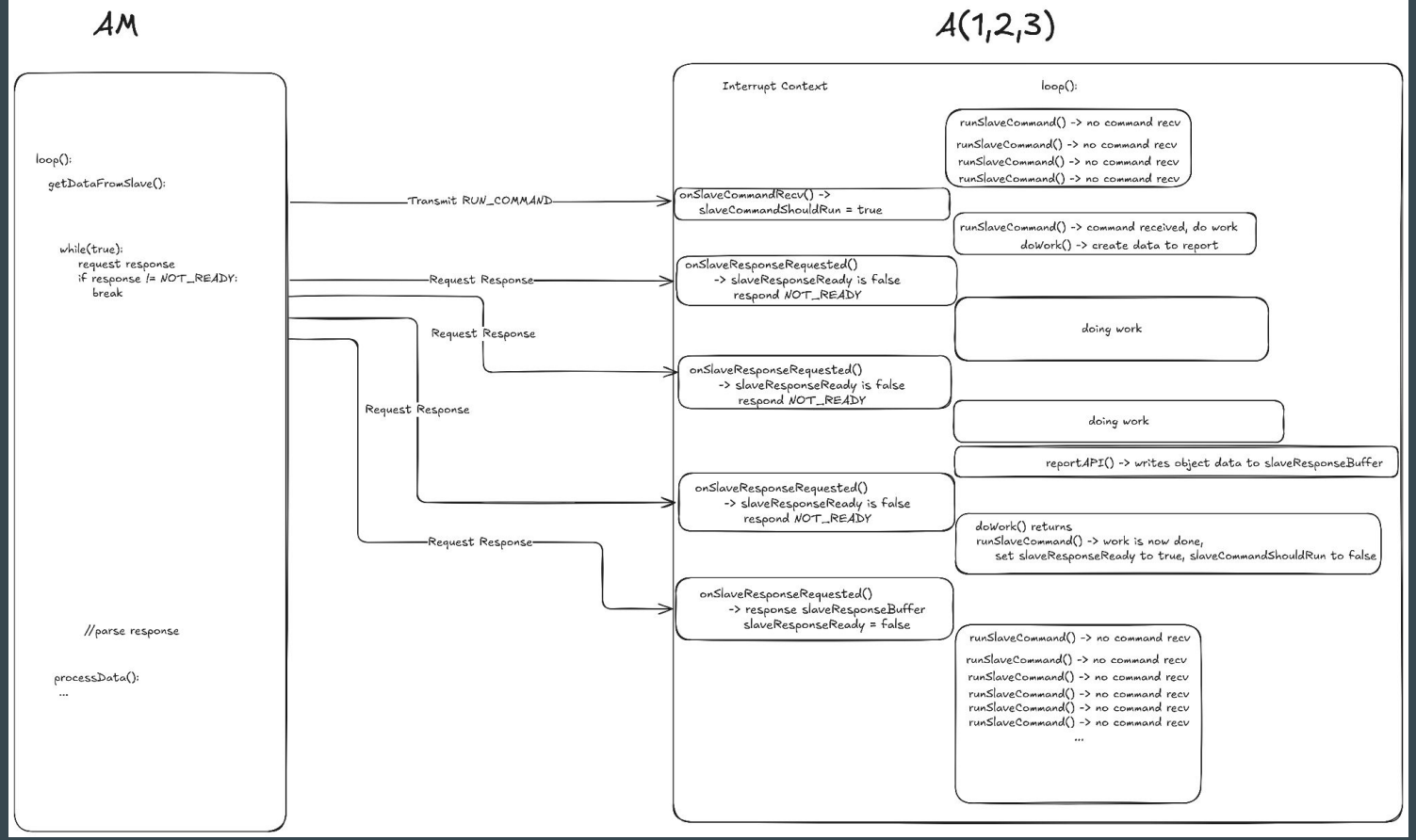
A2

General



Arduino Communication

- Serial
- I2C (SDA/SCL)
- Custom blocking API for sending and receiving 4 byte messages



Arduino Communication - API for synchronization

Roles & Processes

Team Roles

- Sean: AM, Serial Communication
- Valentina: A3 (input), Documentation
- Dominic: A2 (steering), Motors
- Youssef: A1 (Rear)

Workflow Tools/Methods

- Github (Source control, code review)
- Kanban Board for Project Management
- Weekly Meetings

Original Work

This project comprises original work, as the system architecture and multi-Arduino integration resulted from our group's collaboration rather than copying external sources. Other unique elements such as writing our own serial communication API, the programmable path, and controller module being on device rather than in the form of a remote control proves that our project has original work.

The End

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Thank you